

Appendix F – OEM Documentation

This appendix contains the following OEM documentation:

- Yaskawa V1000 Instruction Manual

APPENDIX F – OEM DOCUMENTATION

FastBack® Model 90E Technical Manual

Wiring Diagrams

This Technical Publication contains the following wiring diagrams:

- Wiring diagram for Yaskawa VFD, Model V1000
- Diffused / Reflective Photoeye (Efector # 0G0035) wiring diagram
- Through-Beam Photoeye (Efector # OJ5130 and # OJ5131)
- Ultrasonic Level Sensor wiring diagram
- Position Transducer wiring diagram

Yaskawa V1000 Wiring Diagram

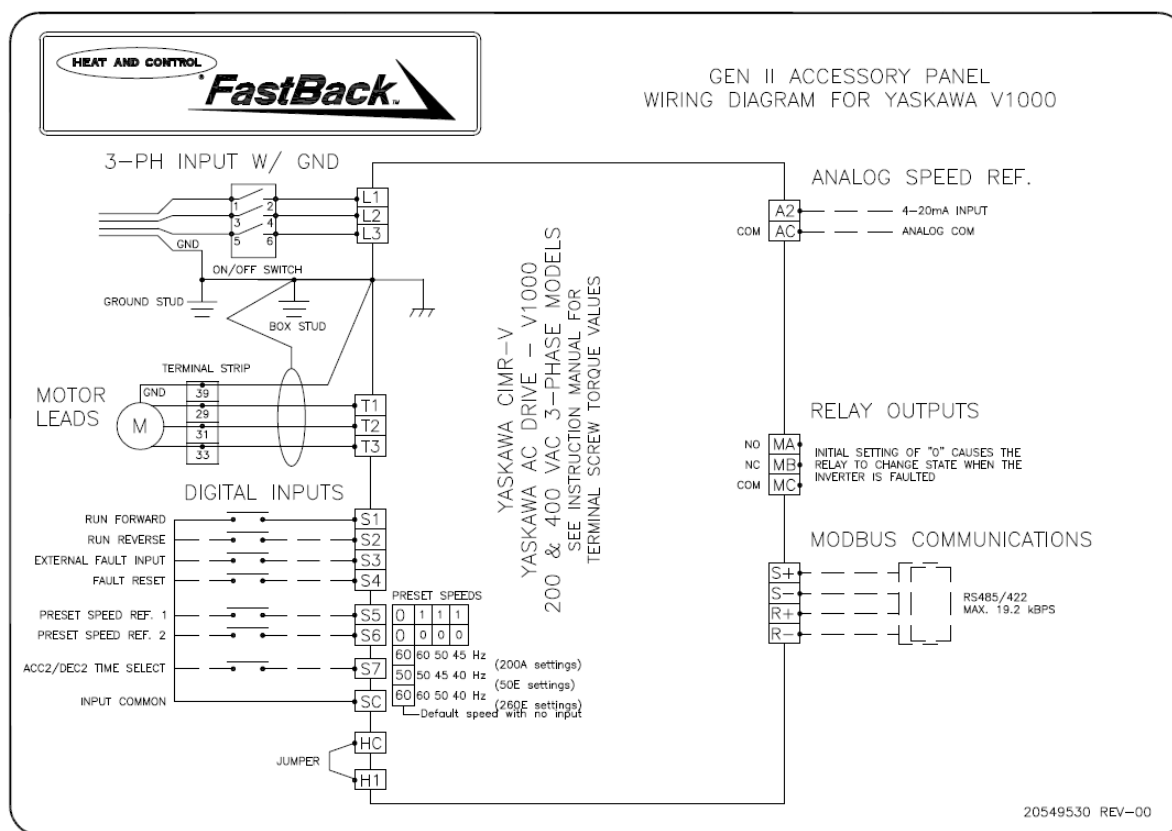


Figure 2 - 1 – Yaskawa V1000 Wiring Diagram

Photoeye Wiring Diagrams

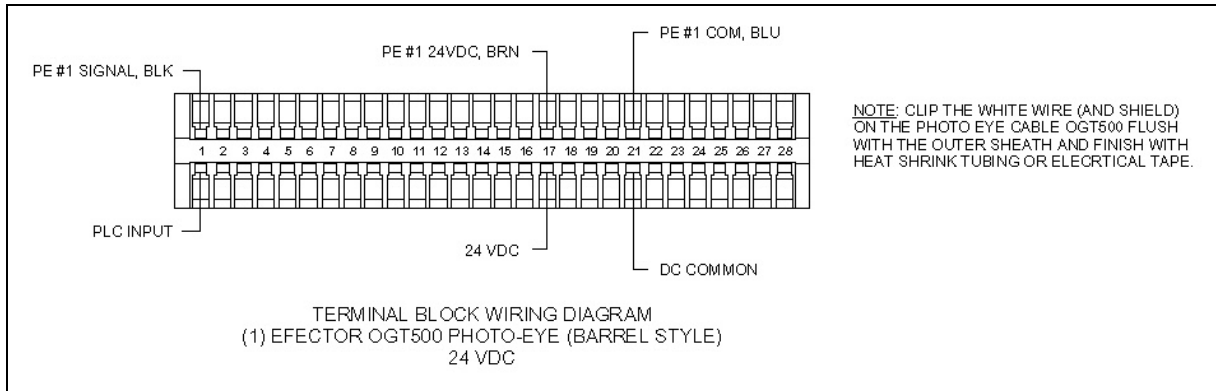


Figure 2 - 2 - Wiring configuration for one diffused / reflective Photoeye

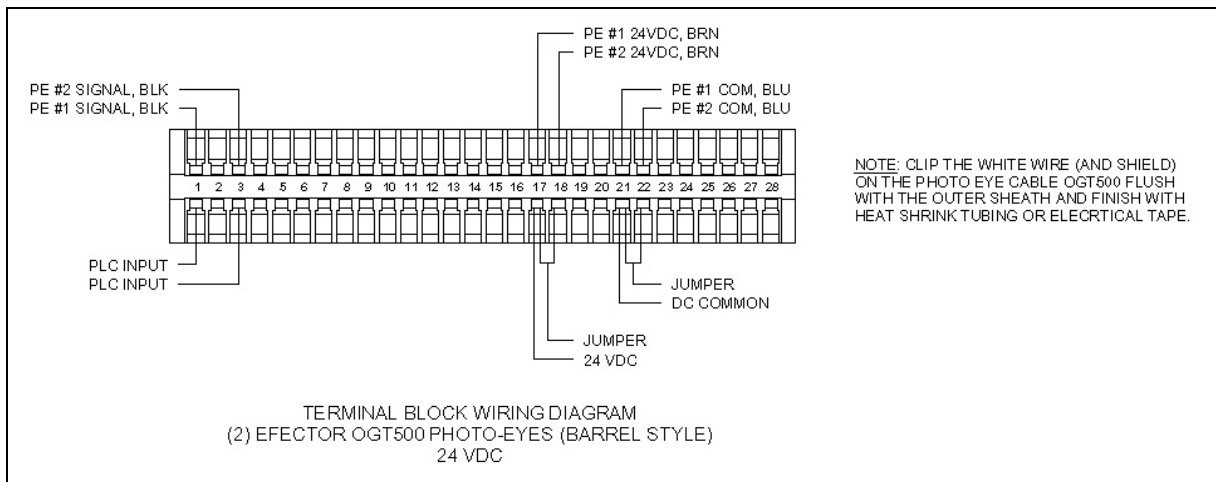


Figure 2 - 3 - Wiring configuration for two diffused / reflective Photoeyes

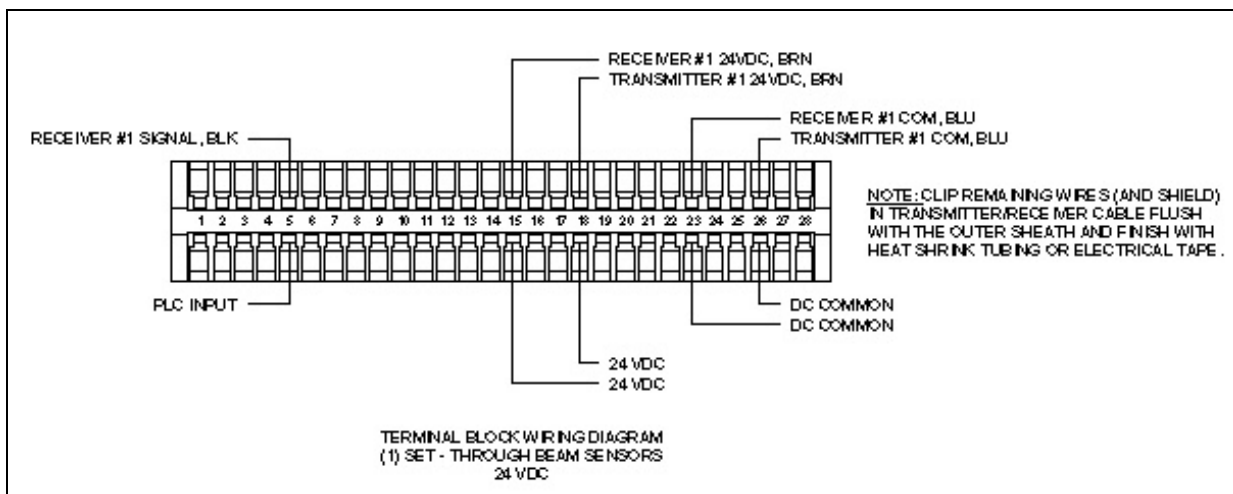


Figure 2 - 4 – Wiring diagram for single through-beam Photoeye

Revolution Gate Wiring Diagram

The following diagram shows the terminations for the typical devices associated with a Revolution Gate: one (1) ultrasonic sensor and one (1) position cable transducer.

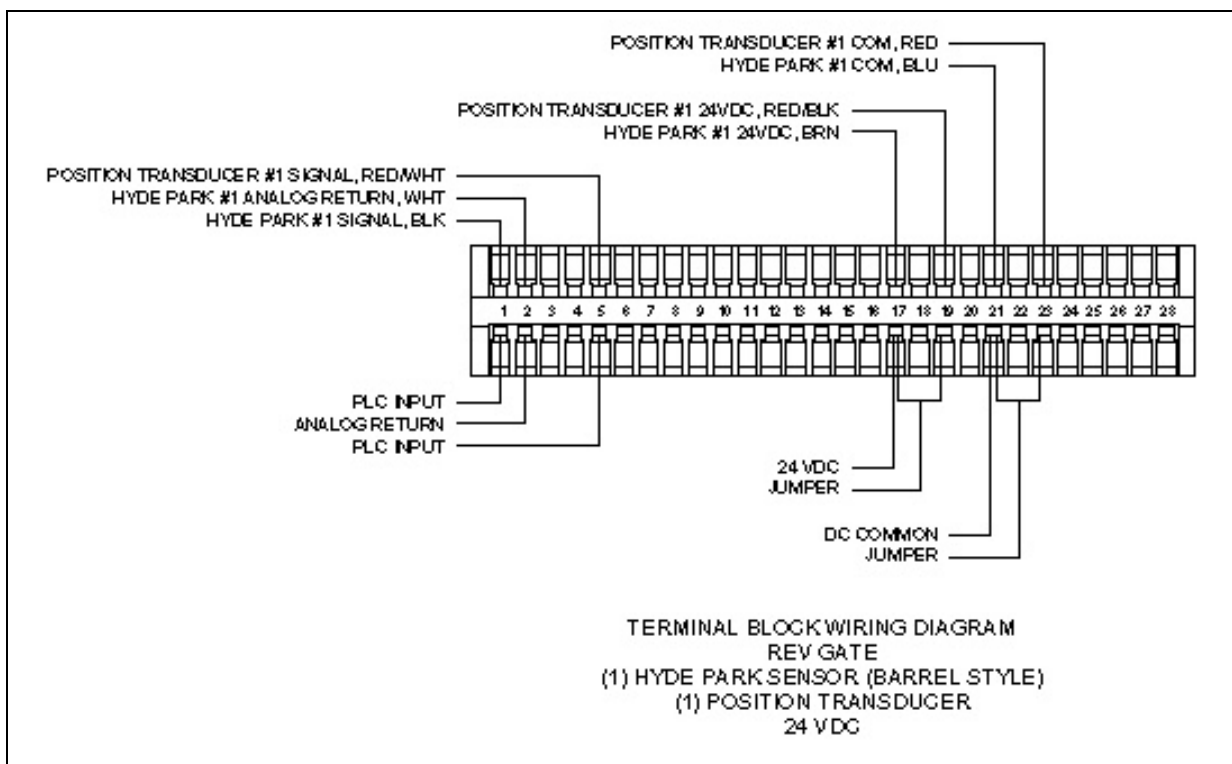


Figure 2 - 8 – Typical Revolution Gate Wiring Diagram including Position Transducer & Ultrasonic Sensor

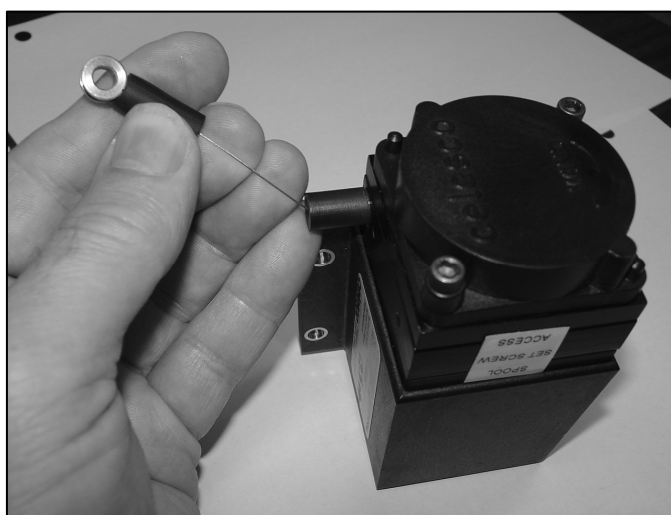


Photo 2 - 1 – Celesco PT1MA Transducer

Component Specifications

This technical publication contains the following component specifications:

Ultrasonic Level Sensor for Product Bed Sensing prior to Revolution Gate:

- Hyde Park Superprox (SM956A-132600) – Barrel Type – With 2500 ms response time

Ultrasonic Level Sensor for Product Bed Sensing in a Revolution Tumble Drum Seasoning Unit:

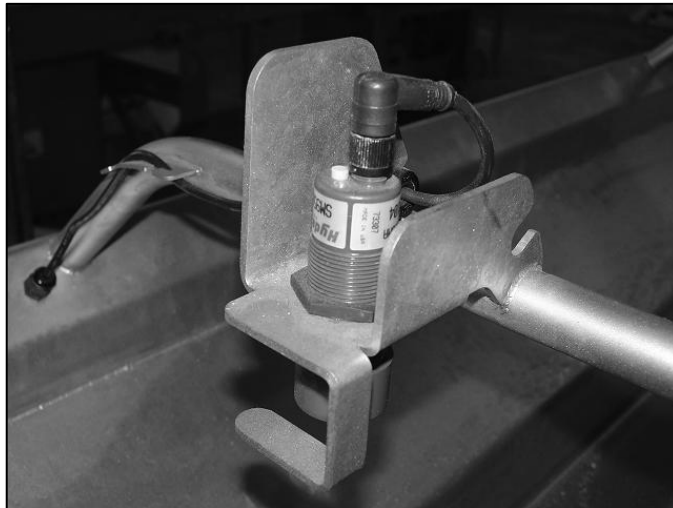
- Hyde Park Superprox (SM956A-132400) – Barrel Type – With 500 ms response time

Setup and Adjustment of the Hyde Park SM956A-132600 and SM956A-132400 Ultrasonic Sensors

The Hyde Park ultrasonic sensor (LV) is used as a level sensor which monitors the product bed depth in a Revolution system pan. Currently two versions of the Hyde Park Ultrasonic Sensor are available. (see Photo 5-1) Externally, these sensors are identical. Internally, the only difference is the response time / time averager value which is set at the supplier's factory.

- The SM956A-132600 sensor has a 2500 ms response time and is the standard sensor used for Revolution Gate systems.
- The SM956A-132400 sensor has a 500 ms response time and is the standard sensor used for Revolution Seasoning System units.

These sensors requires a 24 VDC supply and provides a 4-20 mA output signal proportional to the bed depth of the product in the FastBack pan. This section will briefly describe the basic controls and setup of the sensor. The calibration of the sensor is covered in the Heat and Control Seasoning Equipment Manual under Section 4, Setup & Calibration for OMS / RD Systems and Section 5, Setup and Calibration For Direct Control (DC) Systems. The factory instructions for the sensors can be found in Appendix C, Component Specifications.



*Photo 5 - 1 – Hyde Park ultrasonic sensors SM956A-132600 and SM956A-132400
Both barrel types*

There is only one calibration point for the Hyde Park ultrasonic sensors. They are defined as follows:

Limits Button – The Limits Button sets the sensing range. Press once, with the pan completely empty, to set low level. Then place a piece of cardboard across the pan at the desired upper limit. Press the limits button a 2nd time to set the high level.

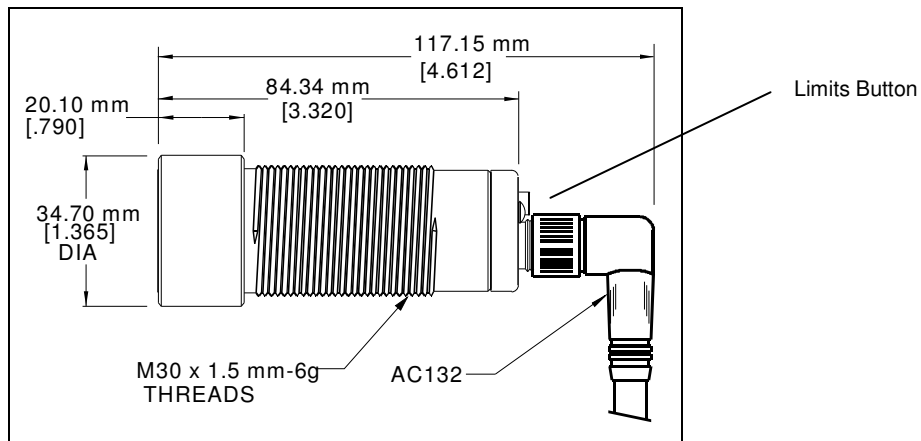


Figure 5 -1 – Exterior Controls of the Hyde Park Sensor

Sonic Profile of Sensor and Height above TDF pan

The ultrasonic sensor has been positioned to allow the maximum sound wave reflection (without false readings from the side walls of the FastBack® pan). (see Figure 5-2)

Note: The Hyde Park Ultrasonic Level Sensor seen in Figure 5-2 represents either sensor offered.

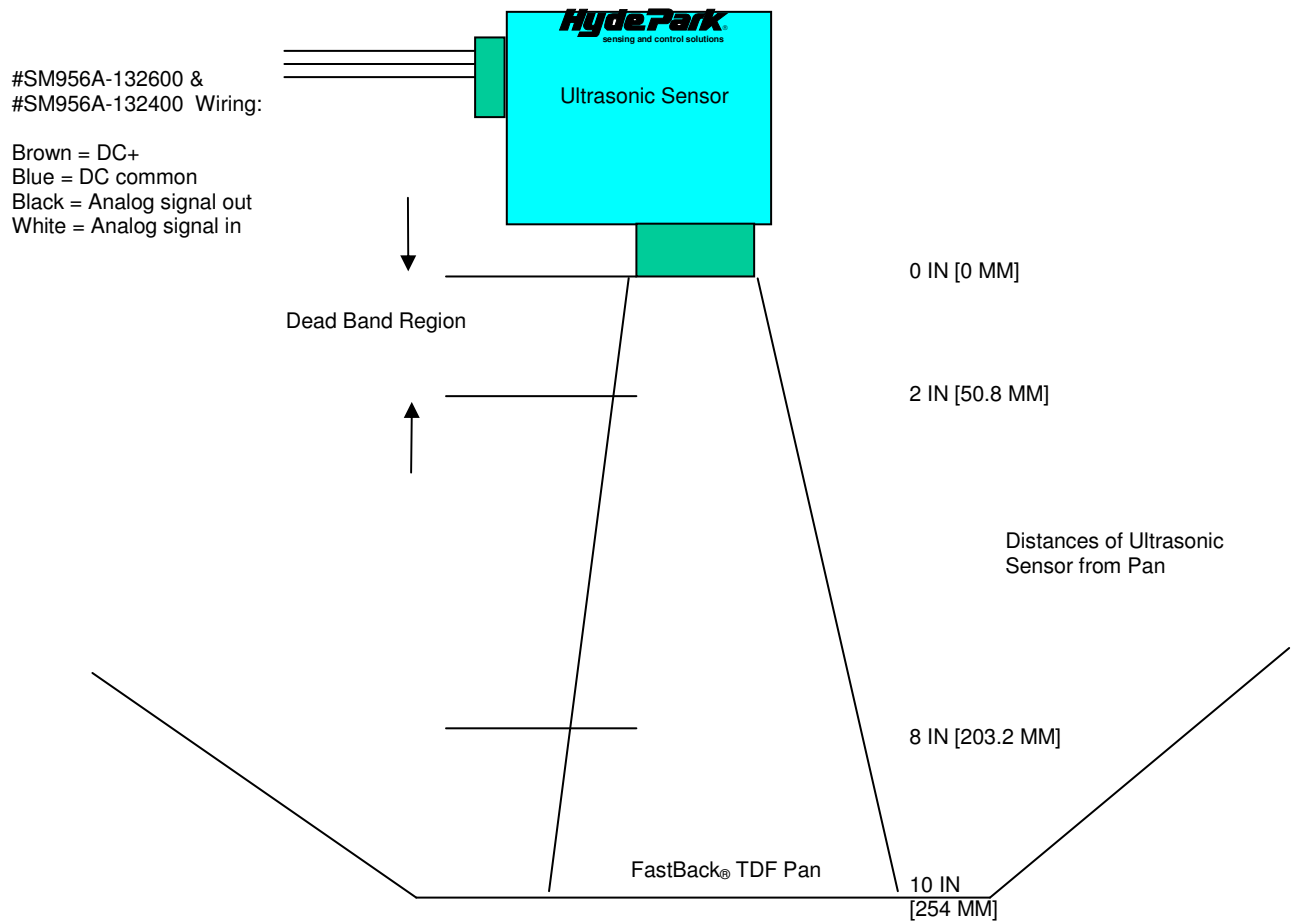


Figure 5 - 2 – Hyde Park ultrasonic sensor placement

Note: There exists a dead band region between 0 and 2 inches [0 and 50.8 mm] from the Hyde Park ultrasonic sensor. Between these distances, the sensor will experience unusable responses.

SUPERPROX®



SM956A-132000

Ultrasonic Analog Output Sensor

30 mm, Direct 4-20 mA Analog Output

Hold output on loss-of-echo, 4 mA at power up

OPERATOR INSTRUCTIONS

This self-contained, ultrasonic analog output sensor provides an analog output signal that is directly proportional to the object position relative to the analog span limits. The analog output is at 4 mA when an object is at or farther than the far analog span limit. The analog output is at 20 mA when an object is at or closer than the near analog span limit. Objects as small as 10 mm (3/8 inch), transparent, opaque, plastic, glass, metal, liquid, or solid can be detected within the sensing range. A multicolor LED indicates the zone of the object and a red LED indicates the magnitude of the 4-20 mA analog output.

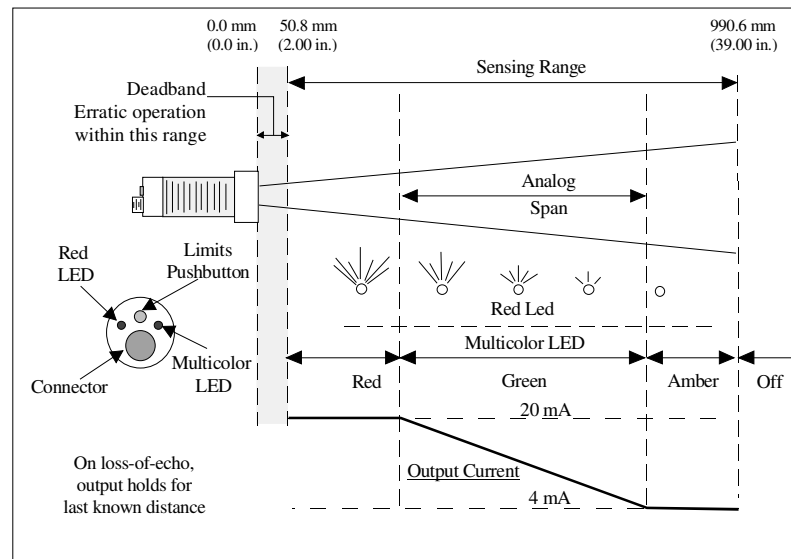
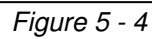


Figure 5 - 3

Literature and application engineering assistance is provided by Hyde Park and its authorized distributors to aid the customer in selecting the product for an application. The customer, however, is responsible for determining the suitability of the product in the application.

Heat and Control Product Handling Systems

Dimensions



Brn + 15 to 24 VDC

Blk Analog signal

Wht Analog return

Blu DC Com

Sensor Connector Pinout
(view of sensor)

2 Analog return White

1 + 15 to 24 VDC Brown

3 DC Com Blue

4 Analog Signal Black

NOTE:
The load must be connected between the analog signal and the analog return. Do not connect the analog return to the other grounds or chassis grounds.

Figure 5 - 5

Operation

The analog output varies directly proportional with the object position relative to the analog span limits. When the object is at or closer than the near analog span limit and farther than the deadband, the analog output is at 20 mA and the red LED is at full brightness. When the object is at or farther than the far analog span limit and within the sensing range, the analog output is at 4 mA and the red LED is off.

Multicolor LED States

The multicolor LED indicates the position of the object

Multicolor Led	Object Position
Red	At or closer than near span limit
Green	Between near and far span limits
Amber	At or farther than far span limit
Off	Object not being detected

Table 5-1

Red LED

The red LED intensity varies directly with the magnitude of the analog output, with off indicating the analog output is at 4 mA and full brightness indicating the analog output is at 20 mA.

Loss of Echo Operation

On loss-of-echo, the multicolor LED is off and the output holds for the last known distance.

No Echo Suppression

This model sensor processes only the first echo return received between the deadband and the sensing range.

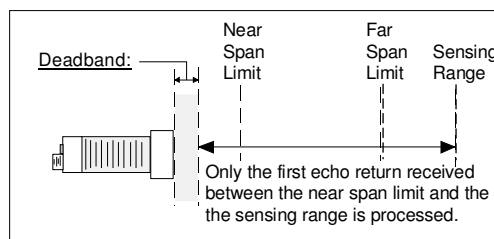


Figure 5 - 6

Setting the Analog Span Limits

Depress the SETUP pushbutton (the multicolor LED rapidly flashes amber to indicate the pushbutton is pressed) until the multicolor LED flashes green (about 3 seconds), and then release the SETUP pushbutton. The multicolor LED continues flashing green indicating the sensor is waiting for the first limit. Align a flat object parallel to the sensor face at the desired distance position for either window limit, and press the SETUP pushbutton once. Upon release of the SETUP pushbutton, the multicolor LED flashes amber indicating the first limit is set and the sensor is waiting for the second limit. Align a flat object parallel to the sensor face at the desired position for the second window limit and press the SETUP pushbutton once. Upon release of the SETUP pushbutton, the multicolor LED turns to the color that indicates where the object is located. The sensor has no timeout for setting limits.

While the SETUP pushbutton is depressed, the multicolor LED turns amber to indicate the sensor detects the object. If the sensor does not detect the object, the multicolor LED is red while the SETUP pushbutton is depressed, and when the SETUP pushbutton is released after not detecting an object, the multicolor LED flashes red 2 seconds, and then requests that limit again by flashing green for the first limit or flashing amber for the second limit.

The limits are saved in nonvolatile memory and thus retained when power is removed from the sensor.

Accessories

Quick Connect Cables

Model: AC130 straight, 4-conductor, 5 meters (16 feet)
 Model: AC132 right-angle, 4 conductor, 5 meters (16 feet)

TECHNICAL PUBLICATIONS PH005 – HYDE PARK SENSOR SETUP & ADJUSTMENT

Heat and Control Product Handling Systems

General Specifications

Power Supply: +15 to 24 VDC @ 80 mA, excluding load

Analog Output: 4-20 mA

Response Time: 25 milliseconds

Loss-Echo Time: Holds last known state on echo loss

Power Up Output State: 4 mA

Operating Temperature:

0°C to 50°C (32°F to 122°F) @ 100% relative humidity

Sensing:

Range: 51 to 991 mm (2.0 to 39.0 inches)

Limit Adjustment Resolution: 0.08 mm (0.003 inch)

Sensor Angle with respect to smooth flat surface: 90° ± 10°

Repeatability: ± 0.86 mm (0.034 inch) from smooth flat surface at
constant air temperature

Quick Disconnect Cables (Optional):

AC130: straight, 4-conductor, 5 meters (16 feet)

AC132: right-angle, 4 conductor, 5 meters (16 feet)

Sensor Housing Material:

Case: GE ULTEM® (Trademark of the General Electric Co.)

Face: FDA approved silicone rubber

Sensor Ratings and Approvals

NEMA 1, 3, 4x, 12, 13, and IP 67

Installation/Overtoltage Category: I

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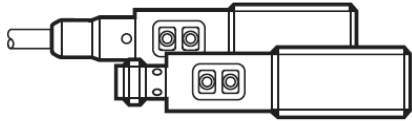
1875 Founders Drive
Dayton, Ohio 45420-4017
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Email: hydepark@hpsensors.com
Web Site: <http://www.hpsensors.com>

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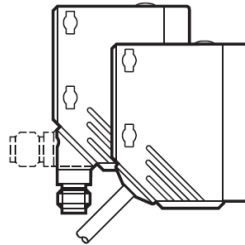
Technical Publication PH006

efector Photoeye Setup

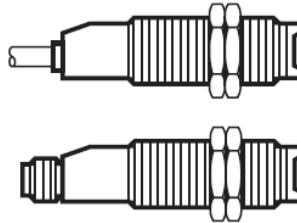
Rev. 02



OGT302
Stainless Steel – Reflective Sensor
Medium Color Range
Max Range: 800 mm



O5H500
Reflective Sensor
Wide Color Range
Max Range: 1800 mm



OG0035
All Purpose Reflective Sensor
Limited to Single Color Detection
Max Range: 600 mm

OJ5130 & OJ5131
Through-Beam Sensor
Max Range: N/A



OID250
Laser Distance Detection Sensor
Wide Color Range
Max Range: 2 M

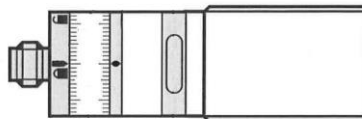


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TECHNICAL PUBLICATION PH006 – EFECTOR PHOTOEYE SETUP

Heat and Control Product Handling Systems

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TECHNICAL PUBLICATION PH006 – EFECTOR PHOTOEYE SETUP

Heat and Control Product Handling Systems

efector Sensors Descriptions & Part Numbers

efector Part #	H&C Part #	Description	Voltage(s)
OGT302	20569969	Diffuse Reflection, SST barrel, 3-Wire	24 VDC <u>only</u>
Primary Cable (#EVT005)	20535083	10-Meter, 4-Wire, Straight Connector	
Alternate Cable (#EVT005)	20548193	10-Meter, 4-Wire, 90° Connector	
O5H500	20535082	Diffused/Reflective, Rectangular body, Visible beam, 3-Wire	24 VDC <u>only</u>
Primary Cable (#EVT005)	20535083	10-Meter, 4-Wire, Straight Connector	
Alternate Cable (#EVT005)	20548193	10-Meter, 4-Wire, 90° Connector	
OG0035	20501821	Diffused/Reflective, Orange barrel, 2-Wire	24 VDC or 110 VAC
Cable (#E18215)	20501824	5-Meter, 3-Pin	
Cable (#E18210)	20516006	10-Meter, 3-Pin	
OJ5130	20527494	Through-beam Transmitter, Visible beam, 2-Wire	24 VDC <u>only</u>
OJ5131	20527490	Through-beam Receiver, Visible beam, 3-Wire	24 VDC <u>only</u>
Cable (#E18227)	20524811	5-Meter, 4-Pin, For both OJ5130 and OJ5131	
OID250	20591952	Photoelectric Sensor, Quick Disconnect, 4-Pin, Class 1 Laser Light	24 VDC only
Cable (#EVT002)	20535083	10-Meter, 4-Wire, Straight Connector	

General Instructions for efector Photoeyes

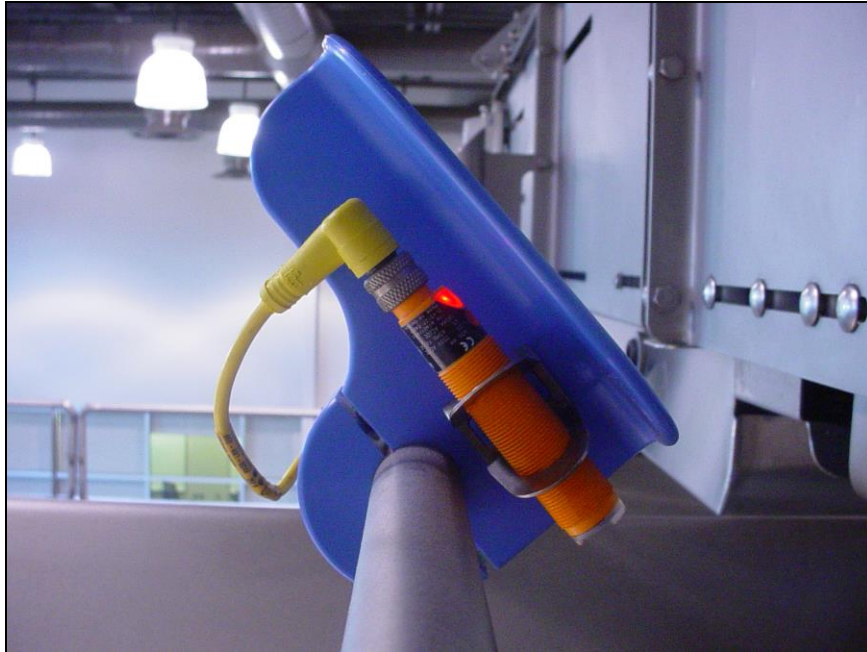


Photo PH006-1 – Photoeye Assembly Positioned at 45° Angle

The *efector* photoeyes are diffuse-reflective photoeyes (they receive the return signal off of the product itself) (see Figure PH006-1). They have a maximum focal length of 600 mm (24 inches) up to 1800 mm (70 inches), depending upon the model. They are all suitable for a washdown environment.

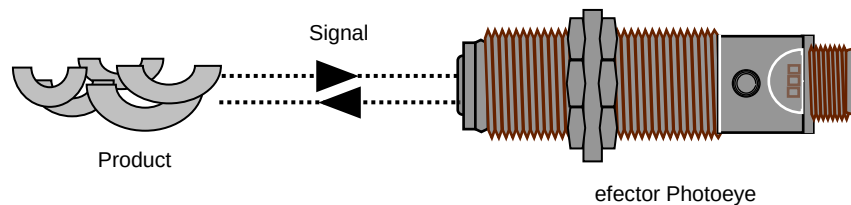


Figure PH006-1 – efector Diffuse Photoeye

TECHNICAL PUBLICATION PH006 – EFECTOR PHOTOEYE SETUP

Heat and Control Product Handling Systems

All FastBack® Models come from the factory with the photoeye assembly attached to the pan assembly (see Photo PH006-2).

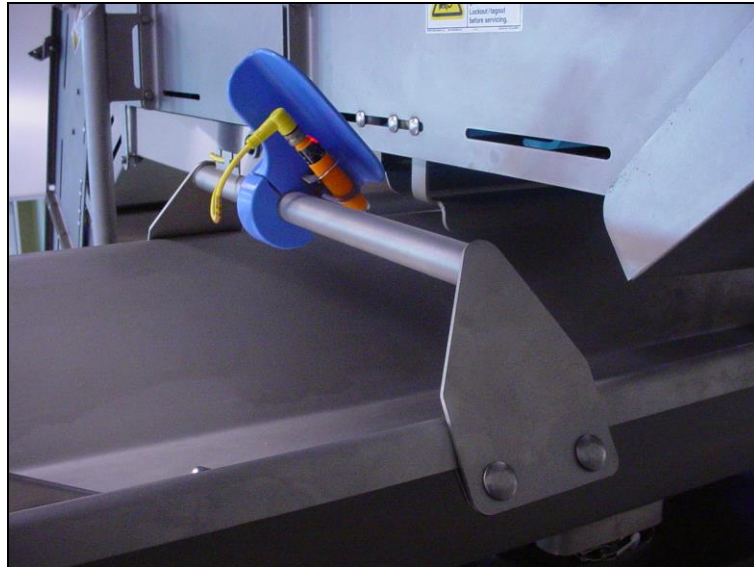


Photo PH006-2 – Photoeye Assembly

A junction box on the drive assembly enclosure connects the photoeye assembly to the control source at your facility. During installation, the photoeye must be wired to its power supply and to the control source (see Photo PH006-3). Refer to the wiring diagram found in each sensor's section. Additionally, the brackets holding the photoeye must be adjusted and tightened so that the photoeye is properly positioned for calibration to the desired bed depth.

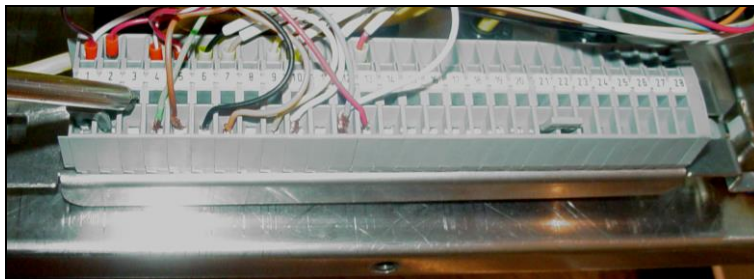


Photo PH006-3 – Wiring of photoeye

Wiring the Photoeye(s)

This publication's sections describe how to connect the photoeye to the primary and control power sources at your facility.



Applies to the efector #OG0035 ONLY, which is a 2-wire sensor.

DO NOT operate the OG0035 directly from a voltage supply without a load. This is a short circuit and may destroy the photoeye.

DO NOT use an incandescent light bulb as a load. If the bulb is cold, the inrush of current will overload the circuit and destroy the photoeye.

Incorrect wiring will destroy the photoeye. If you are not sure of the correct wiring, call the Service Office in Lititz, PA at (800) 648-5313.

For specific wiring termination details, see the Top Level Assembly (TLA) drawings for each piece of equipment. See examples below:

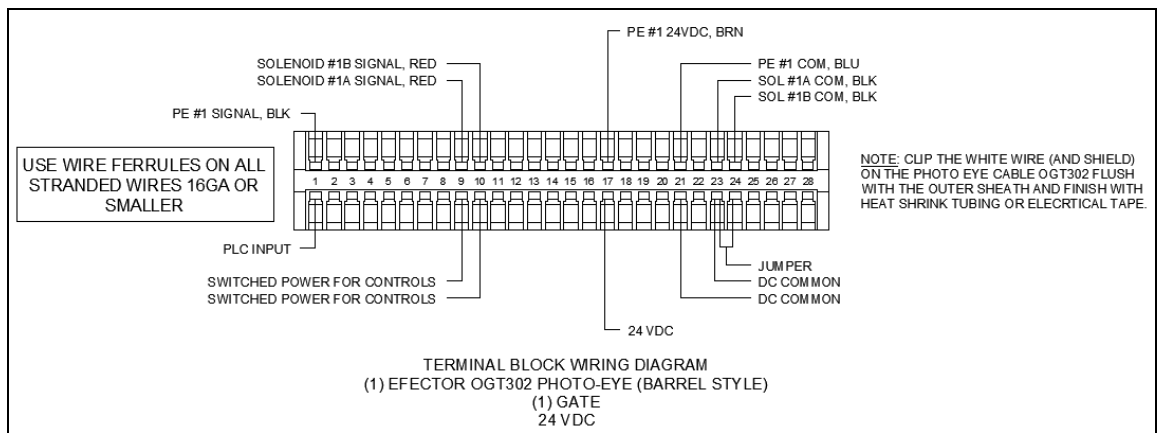


Figure PH006-2 – Typical Fastback 260E-G2 Wiring Diagram

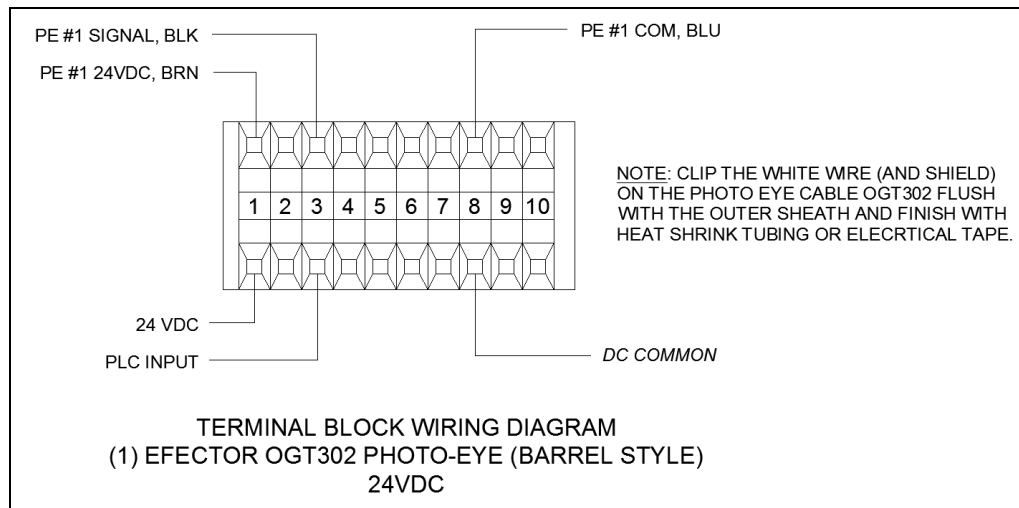


Figure PH006-3 – Typical Fastback 90E-G2 Wiring Diagram

Positioning the Photoeye(s)

The PE is almost always mounted directly to the pan, or a pan cross pipe (see Photo PH006-4). This PE bracket has several methods of articulation.

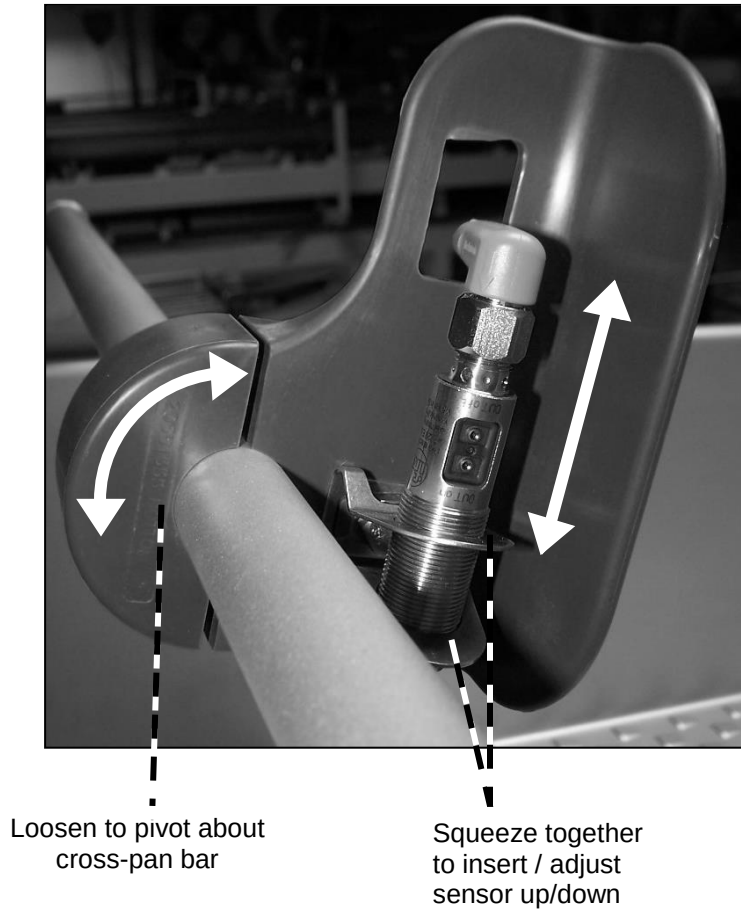
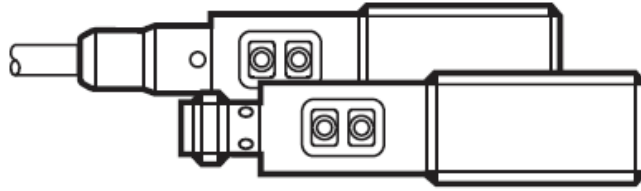


Photo PH006-4 – Typical Barrel-Style Mounting and Articulation Points

Diffused / Reflective Sensor – efector #OGT302 Controls Setup & Line Commissioning (H&C Part #20569969)



Sensor Orientation

Inaccurate readings can result from improper sensor orientation, especially if sensors are positioned vertically, as this can lead to excessive beam bounce-back from the pan bottom. To prevent inaccurate readings, sensors should be angled on their brackets between 10 degrees and 60 degrees (see Figure PH006-4). Typically, the sensor is angled upstream of the product flow, but angling downstream does not inhibit accurate functioning.

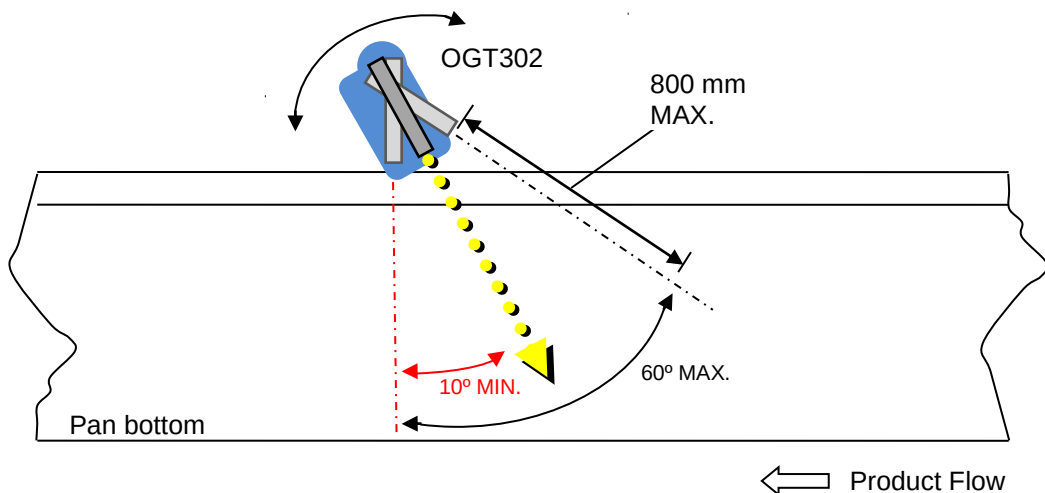


Figure PH006-4 – OGT302 Sensor Orientation

Wiring the #OGT302

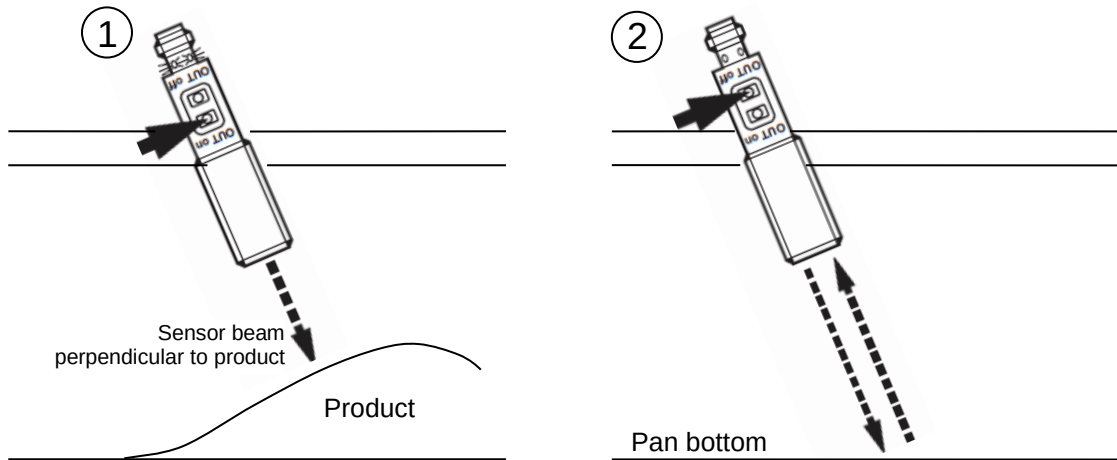


- ▶ Disconnect power.
- ▶ Connect the unit as follows.

	Cable	Connector	Pin connection
L+	BN	pin 1	
L-	BU	pin 3	
load	BK	pin 4	
—	—	pin 2	

Core colours BN = brown, BU = blue, BK = black

Calibrating the #OGT302



- ① Position the product, then press the [OUT on] button. LED flashes.
- ② Remove the product, then press the [OUT off] button. LED goes out.
- ③ Place product (or hand) under sensor to ensure (via LED) it triggers at desired level. Repeat steps 1 and 2 if necessary.

Locking / Unlocking the #OGT302

- Press both buttons for 10 seconds.
- Acknowledgement is indicated by a change of the yellow LED status.

Switching Between Light-On and Dark-On Modes

The OGT302 is typically used in a light-on (light-operate) mode, which is the mode resulting from following the instructions in the “Calibrating the #OGT302” section. In order to switch your OGT302 to dark-on (dark-operate) mode, use the following procedure (opposite of “Calibrating the #OGT302”):

- ① Press the [OUT on] button when no object is present.
- ② Position the product, then press the [OUT off] button.

Sensor Debounce Setup

With the sensor working properly to consistently “see” the product, the last step in commissioning the sensor is to adjust the “Debounce” timers within the system PLC. A Debounce time is the time that the sensor must be continuously made or un-made before its signal is used to do something. This prevents momentary flickering of the sensor from triggering an action. These timers are accessible through the Engineering Screens on the HMI screen. For each sensor, there is a timer value for both the “Covered” and “Uncovered” debounce value. A debounce timer value is typically anywhere from 250 msec (1/4 second) to 2000 msec (2.0 seconds). A reference engineering screen is shown in Figure PH006-5 below.

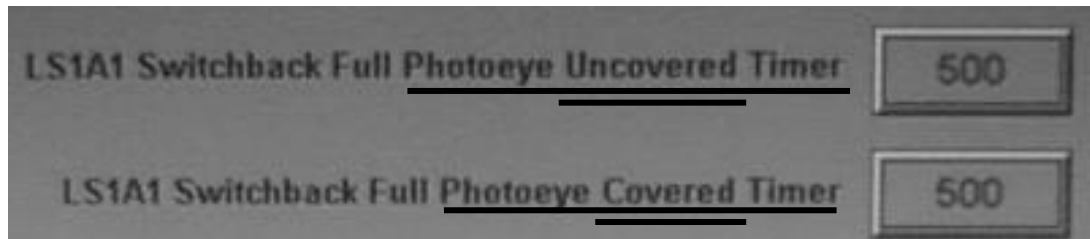


Figure PH006-5 – Engineering screen showing Debounce values (milliseconds)

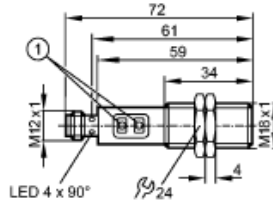
Specifications for the #OGT302

efector200
OGT302



OGTIFPKG/US

Photoelectric sensors



1: Programming buttons

Made in Germany



Product characteristics

Diffuse reflection sensor

Threaded type

M18 x 1

Quick disconnect

Electronic lock

Teach function

Range 2...800mm (Range referred to white paper 200 x 200 mm, 90 % remission)

adjustable

Electrical data

Electrical design	DC PNP
Operating voltage [V]	10...36 DC ¹⁾
Current consumption [mA]	25
Type of light	Infrared light 855 nm
Protection class	II
Reverse polarity protection	yes

Outputs

Output function	light-on / dark-on programmable
Factory setting	light-on mode
Voltage drop [V]	< 2.5
Current rating [mA]	200 (...60 °C) / 150 (...80 °C)
Short-circuit protection	Yes (non-latching)
Overload protection	yes
Switching frequency [Hz]	2000

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Specifications for the #OGT302 (cont.)

efector200
OGT302



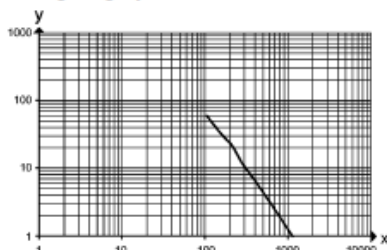
OGTIFPKG/US

Photoelectric sensors

Monitoring range

Light spot diameter [mm] 66 (at maximum range)

Excess gain graphs



Measuring / setting range

Setting range [mm] 150...800

Environment

Ambient temperature [°C] -25...80

Protection IP 65 / IP 67 / IP 68 / IP 69K

Tests / approvals

EMC EN 60947-5-2

Mechanical data

Housing materials stainless steel 316L / 1.4404; PA; LCP; EPDM; TPU (urethane)

Lens material PMMA

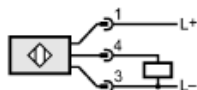
Displays / operating elements

Output status indication LED yellow

Electrical connection

Connection M12 connector

Wiring



Accessories

Accessories (included) 2 lock nuts

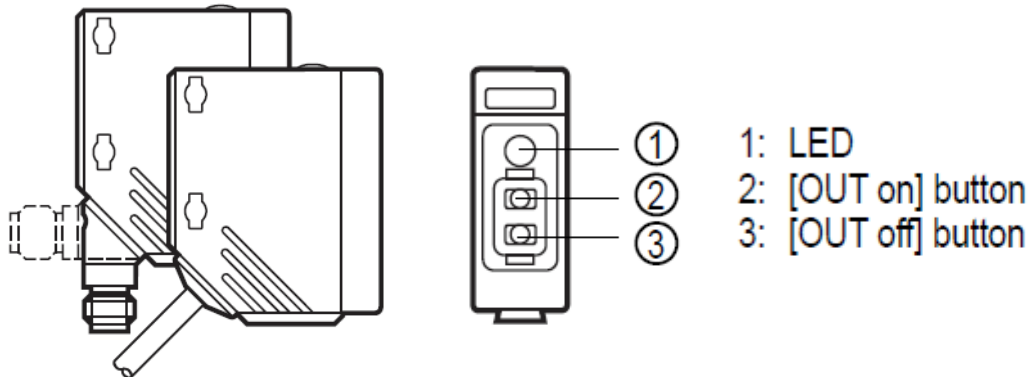
Remarks

Remarks cULus - Class 2 source required

Pack quantity [piece] 1

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Diffused / Reflective Sensor – efector #O5H500 Controls Setup & Line Commissioning (H&C Part #20535082)



Sensor Orientation

Inaccurate readings can result from improper sensor orientation, especially if sensors are positioned vertically, as this can lead to excessive beam bounce-back from the pan bottom. To prevent inaccurate readings, sensors should be angled on their brackets between 10 degrees and 60 degrees (see Figure PH006-6). Typically, the sensor is angled upstream of the product flow, but angling downstream does not inhibit accurate functioning.

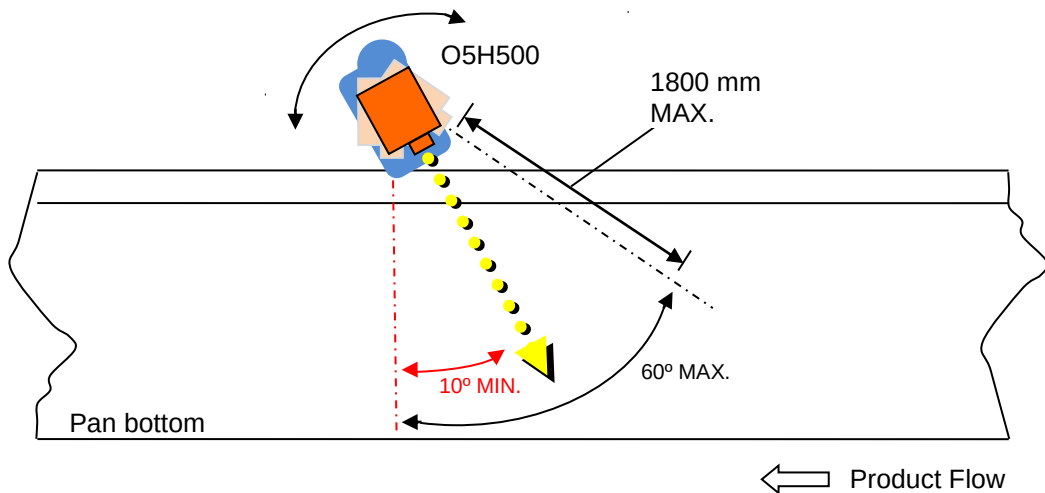


Figure PH006-6 – O5H500 Sensor Orientation

Wiring the #O5H500



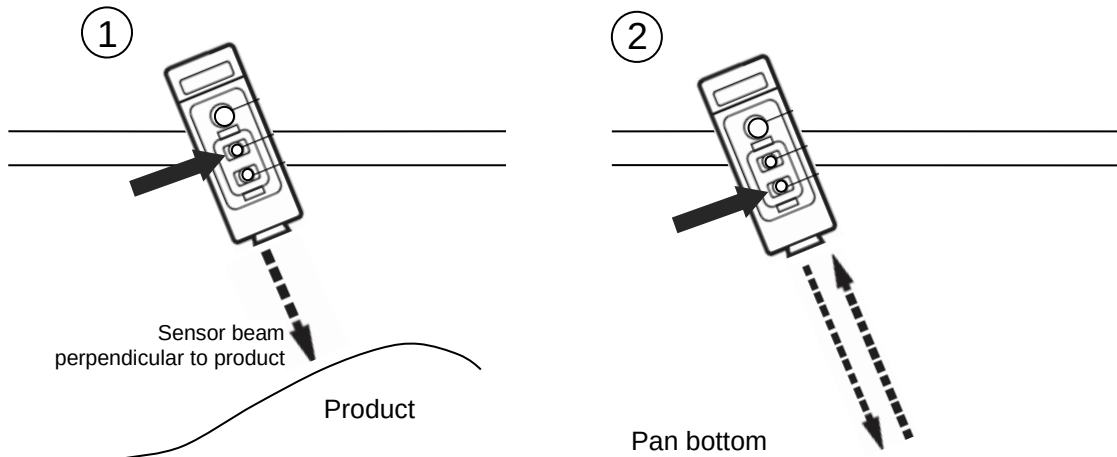
- ▶ Disconnect power.
- ▶ Connect the unit as follows.

	Cable	Connector	Pin connection
L+	BN	pin 1	
L-	BU	pin 3	
load	BK	pin 4	
—	—	pin 2	

Core colours BN = brown, BU = blue, BK = black

Calibrating the #O5H500

NOTE: For this sensor, do **NOT** use pen tip or other sharp object to press calibration buttons. Use blunt instrument instead (eg. Allen wrench).



- ① Position the product, then press the [OUT on] button. LED flashes.
- ② Remove the product, then press the [OUT off] button. LED goes out.
- ③ Place product (or hand) under sensor to ensure (via LED) it triggers at desired level. Repeat steps 1 and 2 if necessary.

Locking / Unlocking the #O5H500

- Press both buttons for 10 seconds.
- Acknowledgement is indicated by a change of the LED status.

Switching Between Light-On and Dark-On Modes

The O5H500 is typically used in a light-on (light-operate) mode, which is the mode resulting from following the instructions in the “Calibrating the #O5H500” section. In order to switch your O5H500 to dark-on (dark-operate) modes, use the following procedure (opposite of “Calibrating the #O5H500”):

NOTE: *For this sensor, do **NOT** use pen tip or other sharp object to press calibration buttons. Use blunt instrument instead (eg. Allen wrench).*

- ① Press the [OUT on] button when no object is present.
- ② Position the product, then press the [OUT off] button.

Sensor Debounce Setup

With the sensor working properly to consistently “see” the product, the last step in commissioning the sensor is to adjust the “Debounce” timers within the system PLC. A Debounce time is the time that the sensor must be continuously made or un-made before its signal is used to do something. This prevents momentary flickering of the sensor from triggering an action. These timers are accessible through the Engineering Screens on the HMI screen. For each sensor, there is a timer value for both the “Covered” and “Uncovered” debounce value. A debounce timer value is typically anywhere from 250 msec (1/4 second) to 2000 msec (2.0 seconds). A reference engineering screen is shown in Figure PH006-7 below.

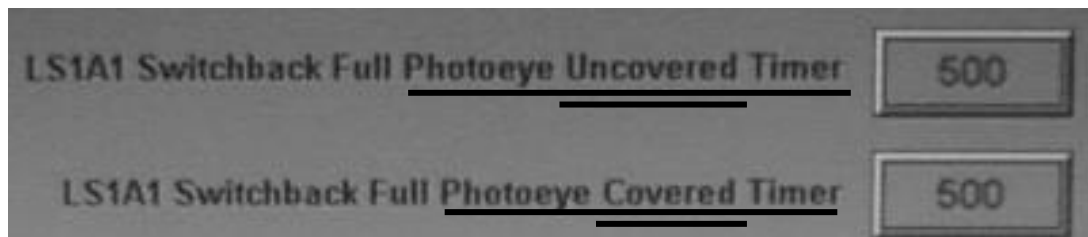


Figure PH006-7 – Engineering screen showing Debounce values (milliseconds)

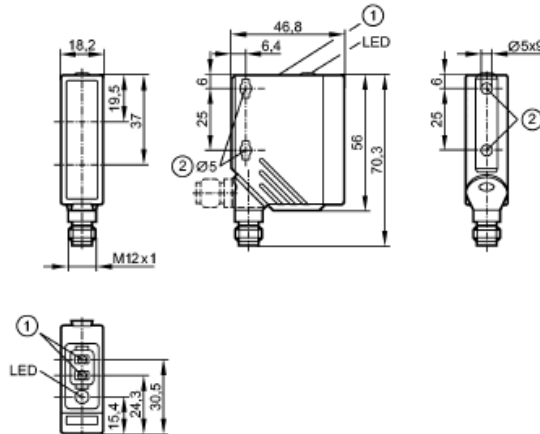
Specifications for the #O5H500

efector200



O5H500

Photoelectric sensors



1: Programming buttons

2: When a M5 mounting screw is used, the max. tightening torque is 2 Nm.

Receiver in upper lens, transmitter in lower lens



Made in Germany

Product characteristics

Diffuse reflection sensor
Rectangular, plastics
Quick disconnect
Background suppression
Teach function
Electronic lock
Range 50...1800mm (Range referred to white paper 200 x 200 mm, 90 % remission)
adjustable

Electrical data

Electrical design	DC PNP
Operating voltage [V]	10...36 DC
Current consumption [mA]	25
Type of light	red light 624 nm
Protection class	II
Reverse polarity protection	yes

Outputs

Output function	light-on / dark-on programmable
Voltage drop [V]	< 2.5
Current rating [mA]	200
Short-circuit protection	Yes (non-latching)
Overload protection	yes
Switching frequency [Hz]	1000

Monitoring range

Range [mm]	
- Object white (90 % remission)	50...1800
- Object grey (18 % remission)	50...900

Specifications for the #O5H500 (cont.)

efector200

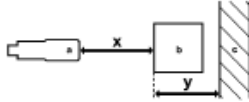


O5H500

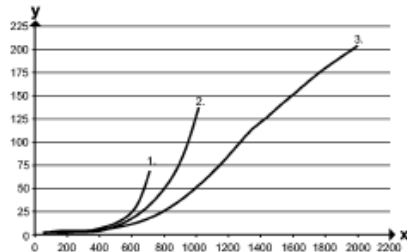
Photoelectric sensors

- Object black (6 % remission)	50...600
Light spot diameter [mm]	50 (Range 1000 mm)

Background suppression chart



- a: sensor
b: object
c: background
x: distance sensor / object
y: min. distance object / background

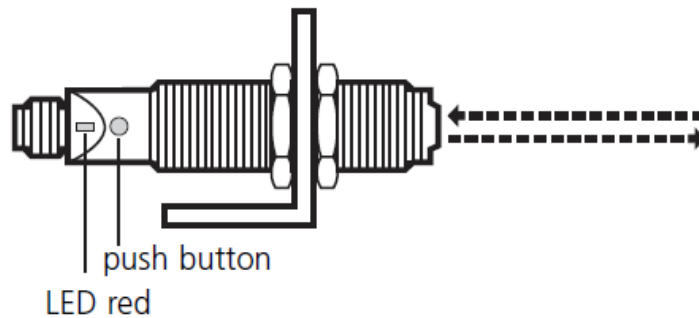


Values in [mm]

- 1 = object black (6 % remission), background white (90 % remission)
2 = object gray (18 % remission), background white (90 % remission)
3 = object white (90 % remission), background white (90 % remission)

Environment	
Ambient temperature [°C]	-25...60
Protection	IP 67
Tests / approvals	
EMC	EN 60947-5-2
MTTF [Years]	360
Mechanical data	
Housing materials	housing: PA; Front framework: high-grade stainless steel; operator interface: TPU (urethane)
Lens material	PMMA
Weight [kg]	0.066
Displays / operating elements	
Output status indication LED	yellow
Electrical connection	
Connection	M12 connector
Wiring	

Diffused / Reflective Sensor – efector #OG0035 Controls Setup & Line Commissioning (H&C Part #20501821)



Sensor Orientation

Inaccurate readings can result from improper sensor orientation, especially if sensors are positioned vertically, as this can lead to excessive beam bounce-back from the pan bottom. To prevent inaccurate readings, sensors should be angled on their brackets between 10 degrees and 60 degrees (see Figure PH006-8). Typically, the sensor is angled upstream of the product flow, but angling downstream does not inhibit accurate functioning.

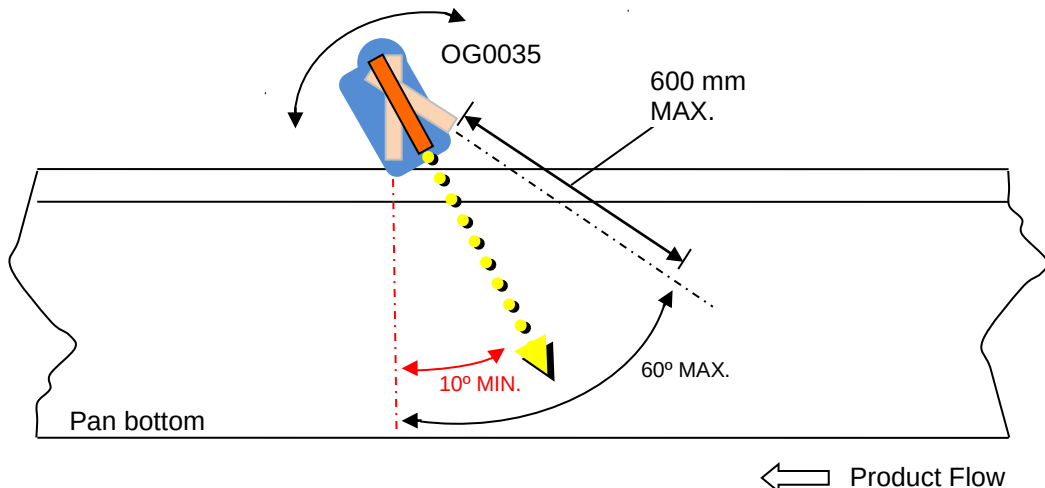


Figure PH006-8 – OG0035 Sensor Orientation

Wiring the #OG0035

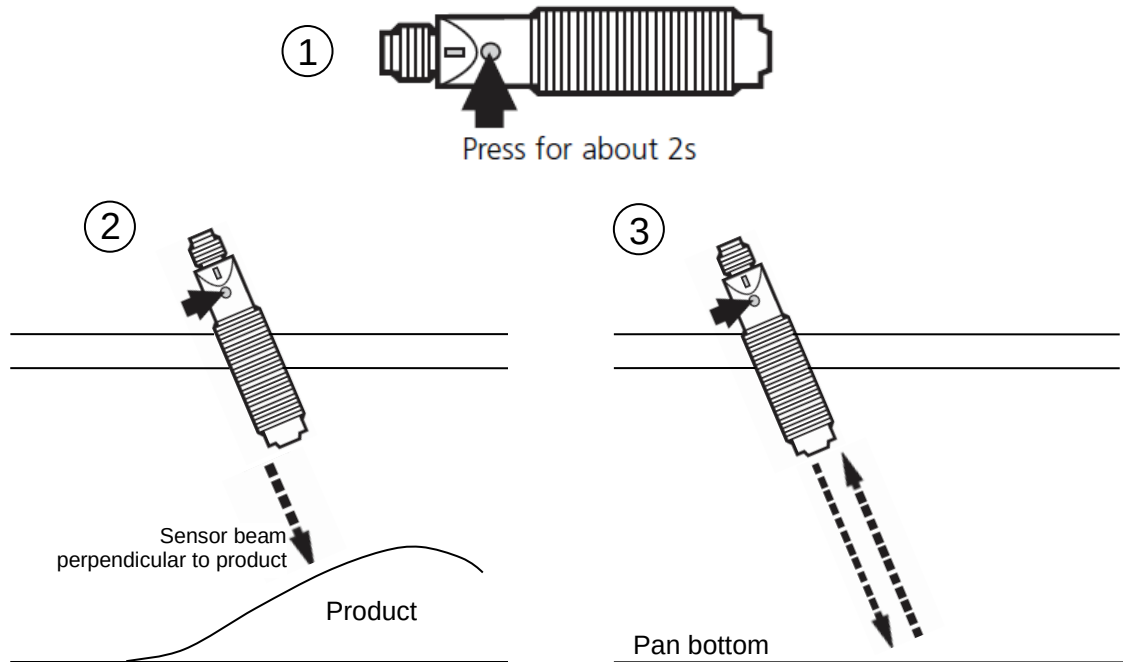


- ▶ Disconnect power.
- ▶ Connect the unit as follows.

Core colours: BN = brown, BU = blue.

Output function light-on (— —)

Calibrating the #OG0035



- ① Press button for 2 s until LED flashes. Programming mode activated.
- ② Position the product, then press button. LED flashes.
- ③ Remove the product, then press button. LED goes out. Normal operation resumes.
- ④ Place product (or hand) under sensor to ensure (via LED) it triggers at desired level. Repeat steps 1 - 3 if necessary.

Locking / Unlocking the #OG0035

- Press button for 10 seconds. LED flashes, first rapidly then slowly.
- LED goes out, indicating the OG0035 has been successfully locked / unlocked.

Sensor Debounce Setup

With the sensor working properly to consistently “see” the product, the last step in commissioning the sensor is to adjust the “Debounce” timers within the system PLC. A Debounce time is the time that the sensor must be continuously made or un-made before its signal is used to do something. This prevents momentary flickering of the sensor from triggering an action. These timers are accessible through the Engineering Screens on the HMI screen. For each sensor, there is a timer value for both the “Covered” and “Uncovered” debounce value. A debounce timer value is typically anywhere from 250 msec (1/4 second) to 2000 msec (2.0 seconds). A reference engineering screen is shown in Figure PH006-9 below.

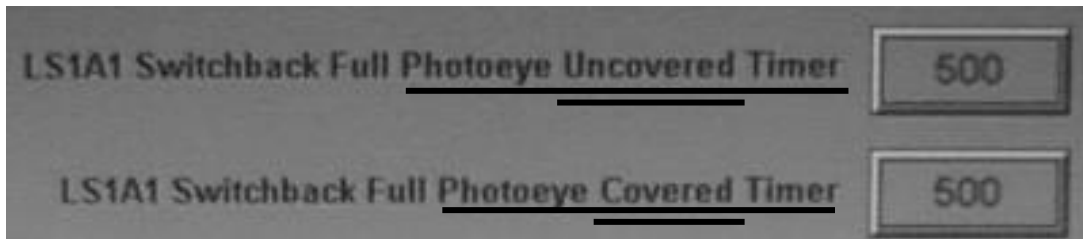


Figure PH006-9 – Engineering screen showing Debounce values (milliseconds)

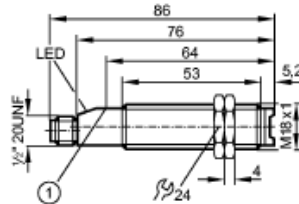
Specifications for the #OG0035

efector200



OG0035

Photoelectric sensors



1: pushbutton



Made in Germany

Product characteristics

Diffuse reflection sensor

Threaded type

M18 x 1

Quick disconnect

Range 1...600mm (Range referred to white paper 200 x 200 mm, 90 % remission)

adjustable

Electrical data

Electrical design

AC/DC

Operating voltage [V]

20...250 AC/DC (47...63 Hz AC)

Type of light

Infrared light 880 nm

Protection class

II

Reverse polarity protection

no

Outputs

Output function

light-on mode

Voltage drop [V]

< 8

Minimum load current [mA]

5

Leakage current [mA]

< 1.7

Current rating

- Current rating (continuous) [mA]

180 (...40 °C) / 90 (...80 °C)

- Current rating (peak) [mA]

î: 0.6 A (20 ms / 0.5 Hz)

Short-circuit proof

no

Overload protection

no

Switching frequency [Hz]

25

Monitoring range

Light spot diameter [mm]

< 169 (at maximum range)

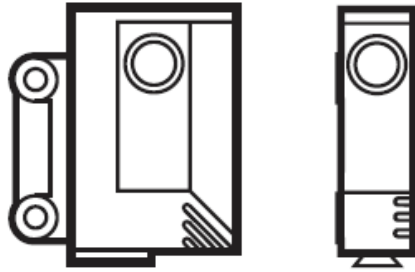
Excess gain graphs

Specifications for the #OG0035 (cont.)

Environment		
Ambient temperature	[°C]	-25...80
Protection		IP 67
Tests / approvals		
EMC	EN 60947-5-2 EN 55011:	class B
MTTF	[Years]	338
Mechanical data		
Housing materials		PBT
Lens material		PMMA
Weight	[kg]	0.046
Displays / operating elements		
Output status indication	LED	red (permanently lit)
Function display	LED	red (flashing)
Electrical connection		
Connection		1/2" connector
Wiring  Note: miniature fuse to IEC60127-2 sheet 1, ≤ 2 A (fast acting)		
Accessories		
Accessories (included)		2 lock nuts
Remarks		
Remarks		Recommendation: check the unit for reliable function after a short circuit.
Pack quantity	[piece]	1

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Through-Beam Sensors – efector #OJ5130
(H&C Part #20527494) Transmitter & efector #OJ5131
(H&C Part #20527490) Receiver Controls Setup & Line
Commissioning



Sensor Orientation

To prevent inaccurate readings, transmitter and receiver sensors should be aligned properly (see Figure PH006-10). The heights of the transmitter and receiver can also be adjusted as needed for varying product bed-depths.

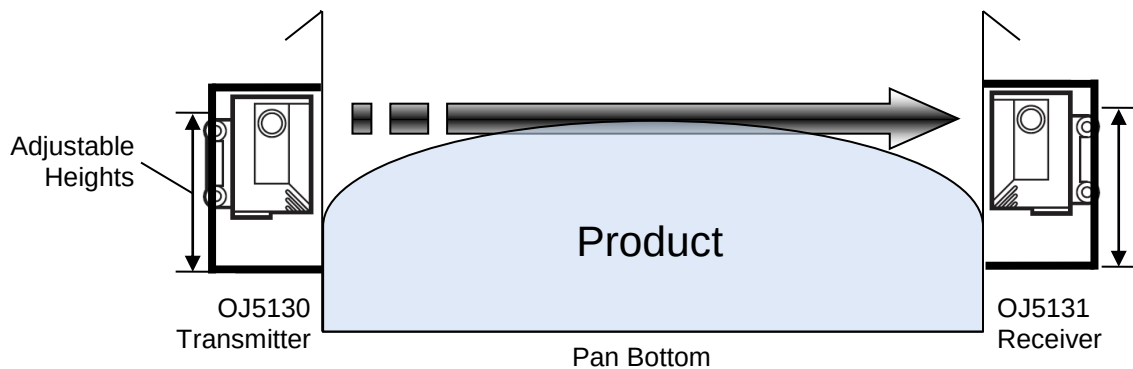


Figure PH006-10 – OJ5130 / OJ5131 Sensor Orientation

Often, through-beam sensors are stationed in pairs to provide variable bed-depth levels or to enable different actions in response to each sensor's trigger (see Photo PH006-5). When this set-up is chosen, transmitters (OJ5130) and receivers (OJ5131) are alternated so as to prevent interfering chatter between the two signals, which is seen in the end-view of Figure PH006-11.

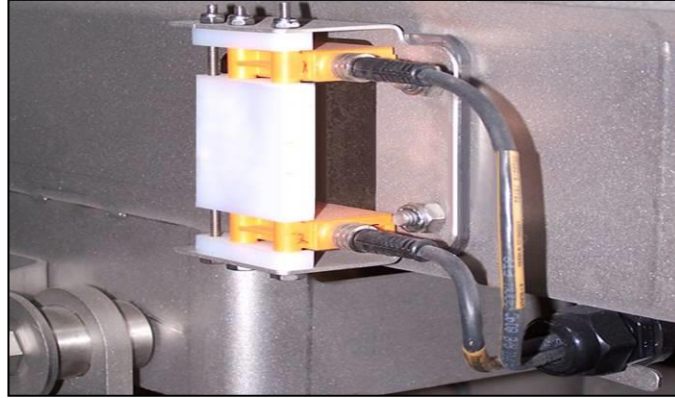


Photo PH006-5 – OJ5130 and OJ5131 Installed

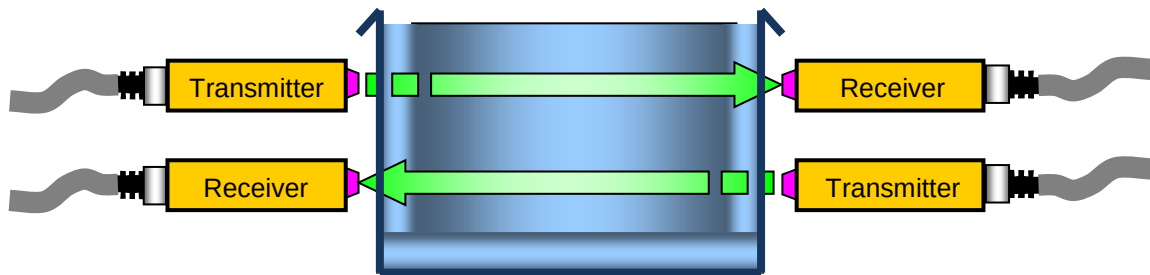


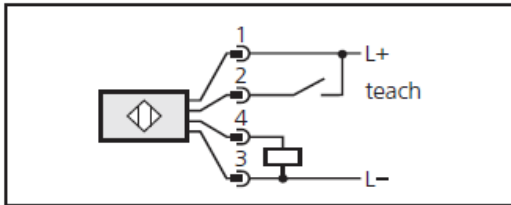
Figure PH006-11 – End view of alternating positions of transmitters and receivers

Wiring the #OJ5130 / #OJ5131



- ▶ Disconnect power.
- ▶ Connect the unit as follows.

Receiver DC PNP

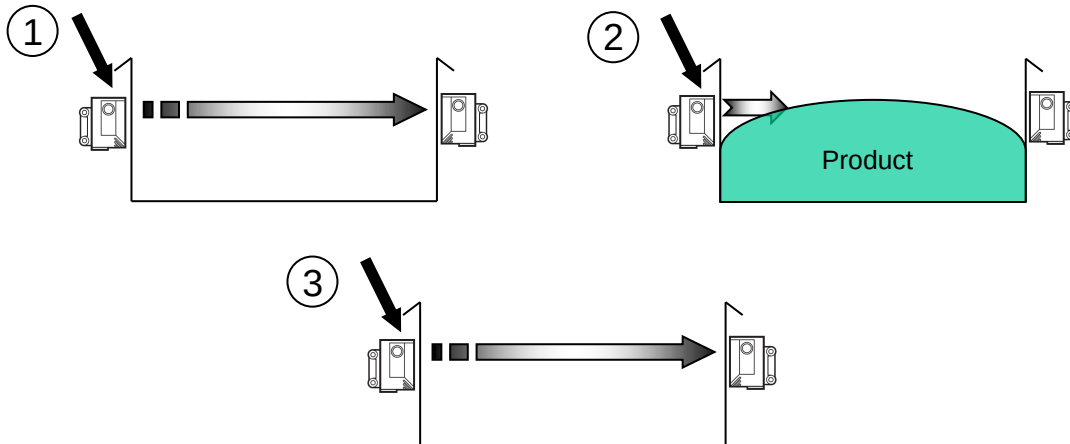


Transmitter DC



Core colours: BN = brown, BU = blue, BK = black, WH = white.

Calibrating the #OJ5130 / #OJ5131



- 1 Push button for 2 s until red LED flashes. Red LED goes out. Yellow and green LEDs flash alternately. Programming mode activated.
- 2 Position the product, then press button. Yellow and green LEDs go out momentarily, then flash again alternately.
- 3 Remove the product, then press button. Yellow and green LEDs go out momentarily, then green LED stays lit. Normal operation mode resumes.
- 4 Place product (or hand) between sensors to ensure (via LED) they trigger when beam is broken. Repeat steps 1 - 3 if necessary.